

Supplementary Methods: reproducibility and evidence attribution

Immune gene and protein profiles across blood, cerebrospinal fluid and peripheral nerve in inflammatory neuropathies

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Purpose and evidence attribution

This appendix documents the statistical procedures and source attribution for the current manuscript. The study combines reanalysis of public transcriptomic resources with synthesis of published CSF protein estimates and localization of published genetic candidates. These components have different inferential units and are identified separately in the figure and table index in Supplementary Data 1.

Supplementary Data 1 retains the preceding aggregate tables and adds gene coverage, donor design summaries, Fc sensitivity tests, refined annotation results, spatial coverage and source access audits. Donor intermediates are retained in the private author materials. Public reconstruction begins from the original source files.

Blood cohorts and cross-cohort synthesis

GSE211225 contains six acute, ten postacute and six control whole-blood profiles. GSE31014 contributes seven GBS and seven control leukocyte profiles. PRJNA1293757 contributes PBMC data from three patients and two controls; monocyte expression is aggregated by biological sample.

Seven script-defined gene modules were evaluated within each blood cohort. Cohort effects used Hedges g and exact label permutation tests of mean score differences. The supplied common.py enumerates all label partitions when their number does not exceed 250000; the acute–control comparisons lie below this limit. Benjamini–Hochberg correction covers seven modules within each comparison. In v39, acute–postacute inferential results are omitted

25 because cross-phase donor independence could not be verified from the archived design.
26 Postacute profiles were included in the original preprocessing.
27 The cross-cohort model uses inverse variance weighting with the DerSimonian-Laird estimate
28 of between-study variance. The reported modified Hartung-Knapp intervals use a residual
29 variance scale bounded below by one and a t distribution with k minus one degrees of
30 freedom. The repository script reproduces the estimates, intervals, P values and I² for all
31 seven modules. Leave-one-cohort-out analyses retain two cohorts and assess direction, with
32 no claim of confirmatory significance.
33 CXCL8 recruitment and complement have positive effects in all three blood cohorts. Four
34 other modules have positive effects in two cohorts. The LIF and OSM module is positive only
35 in whole blood. None of the seven pooled effects reaches P below 0.05.

36 **Peripheral nerve analyses**

37 GSE285983 provides sural nerve single nucleus data. The principal comparison includes 9
38 CIDP and 11 CIAP donors, with eligible donor counts varying by cell group. Counts are
39 summed for each donor and annotated cell group. Groups containing at least 20 nuclei per
40 donor contribute to targeted pseudobulk comparisons. Library sizes use all genes. Targeted
41 expression is transformed as $\log_2(\text{CPM} + 0.5)$.
42 Within each cell group, genes are standardized across eligible donors from all diagnoses.
43 Gene module scores are means of available standardized genes. The original CIDP and CIAP
44 comparisons use the U test of Mann and Whitney with two tails and correction across eight
45 modules within each cell group. These original tests are unadjusted. Version 41 adds the
46 demographic, composition, annotation and broader correction checks described below.
47 Cell fractions are compared at donor level using unadjusted Mann-Whitney tests, with
48 multiple testing correction across cell groups. Spearman correlations among five selected
49 module scores use nine matched CIDP donors. The ten pairwise tests receive one Benjamini-

50 Hochberg correction. The fifteen correlations with INCAT receive a separate correction. No
51 result in either correlation family survives FDR 0.05. Nominal associations remain
52 exploratory.

53 The original Fc score was reproduced from the archived donor expression table, with a
54 maximum absolute score difference below 0.000000000001. Reaggregation of all 20 case and
55 comparator raw matrices reproduced 220 macrophage gene expression values to within the
56 same numerical tolerance. Broader correction across 80 module contrasts gives $q = 0.087$ for
57 the Fc module, compared with $q = 0.0087$ for the original eight module family.

58 **Exploratory Fc analysis specification**

59 The new sensitivity plan was written before the first successful new statistical run, after the
60 original Fc association was already known. It is not a preregistration. An input coverage
61 check identified FCGR2C as absent before new test results were generated. The available
62 gene family therefore contains 11 tests. The original module includes FCGR1A, FCGR2A,
63 FCGR2B, FCGR3A, FCGR3B, FCGRT, FCER1G, TYROBP, SYK, LYN and HCK.

64 **Demographic sensitivity and exact permutation**

65 All nine CIDP donors are male. CIAP includes seven male and four female donors. CIDP
66 counts at Essen, Münster and Würzburg are five, one and three, respectively. Corresponding
67 CIAP counts are one, six and four. Linear models use an intercept, a CIDP indicator, age and
68 indicator variables for centre. The sex sensitivity adds a female indicator. HC3 covariance
69 uses squared residuals divided by the square of one minus leverage inside the sandwich
70 estimator. Confidence limits and tests use a t distribution with residual degrees of freedom.
71 These intervals describe adjusted mean score differences. Mann and Whitney P values are
72 reported for unadjusted comparisons. No effect in female CIDP donors is estimable from this
73 design.

74 **Permutation assumptions and omission analyses**

75 The exact centre permutation test preserves the number of CIDP labels within each centre. It
76 enumerates all 1470 possible allocations and compares the absolute centre adjusted disease
77 coefficient with the observed coefficient. Seven allocations are at least as extreme, giving $P =$
78 0.0047619. The observed allocation is included in the denominator, so no Monte Carlo
79 correction is added. The test assumes exchangeability within centre and cannot remove
80 unmeasured confounding. Separate donor, centre and gene omissions are correlated sensitivity
81 checks. Their nominal P values are descriptive and are not treated as independent replications.

82 **Functional component definitions**

83 Four exploratory summaries distinguish activating receptors FCGR1A, FCGR2A and
84 FCGR3A, the inhibitory receptor FCGR2B, the neonatal Fc receptor FCGR3B, and signalling
85 components FCER1G, TYROBP, SYK, LYN and HCK. Correction covers these four
86 summaries separately. FCGR3B remains in the original module and individual gene table but
87 is outside the four summaries. Gene standardisation uses eligible donors from all diagnoses
88 for these component analyses. Standardisation within CIDP and CIAP alone is a separate
89 sensitivity analysis. Both component families were motivated by the original composite result.

90 **Refined annotation and composition**

91 The original broad macrophage annotation is the union of Macro1 and Macro2 from the
92 whole nerve atlas. Refined immune annotations use Macro1 through Macro18 and additional
93 immune categories. These labels refer to different annotation levels and must not be
94 interchanged. Barcode joins retain all original macrophage nuclei. Thirty two refined
95 categories occur in the focused cohort, including reassigned immune populations and nuclei
96 with no refined label. Fractions use the original macrophage count as denominator. Each
97 category is tested at donor level, with correction across all 32 tests.

98 **Composition model**

99 A pseudocount of 0.5 is added to each donor category count. Log counts are centred within
100 donor to form centred log ratios. Principal component analysis is performed across the 20
101 donors without using disease labels. The first two components explain 22.7% and 20.5% of
102 composition variation and enter the Fc model with age and centre. The estimated disease
103 coefficient is -0.310 , with a 95% interval from -1.038 to 0.419 and $P = 0.375$. This model
104 has 13 residual degrees of freedom. Composition could be a mediator, a confounder or a
105 marker of technical variation. The model does not identify which explanation applies.

106 **Raw matrix annotation checks**

107 The 20 original count matrices are summed by donor under the original macrophage union,
108 original Macro1, original Macro2, the refined macrophage union and each of the 18 refined
109 macrophage subtypes. Library sizes contain all measured genes. Each donor requires at least
110 20 nuclei for the annotation. Tests additionally require at least three eligible donors per
111 diagnosis. Original Macro1 has no eligible donors. Three of four broad annotation definitions
112 and nine of 18 individual refined subtypes are eligible. Broad annotation checks receive
113 correction across three tests, and refined subtype tests receive a separate correction across
114 nine. Genes are standardized across eligible CIDP and CIAP donors within each definition, so
115 effect magnitudes use definition specific scales.

116 **External data measurement and access audit**

117 All eight GSE285984 cell feature matrices contain the same 99 biological genes. None of the
118 specified Fc module genes is present. The two CIDP and two CIAP spatial donors overlap the
119 original nerve atlas, so even covered markers would provide internal triangulation. No spatial
120 distance or expression association analysis was performed. The PRIDE PXD056286 inventory
121 contains 19 instrument files, four Proteome Discoverer files and one checksum file. These are

122 file counts, not participant counts. Its article supplement contains figure descriptions, with no
123 donor protein abundance table. No new quantitative proteomic comparison was performed.

124 **CIDP blood candidates**

125 Kim and colleagues report four KBDS accessions, KSX10000018 through KSX10000021, for
126 a single cell discovery study with two CIDP and two healthy donors. An analysable matrix
127 was not retrieved for this revision. The larger clinical cohort must not be substituted for these
128 single cell donor counts. EGA EGAD00001015725 is a controlled access candidate. The
129 Theranostics T cell study states that data can be obtained from the corresponding author. No
130 data access requests were sent. The audit distinguishes candidate datasets from datasets
131 actually reanalysed and includes the source URLs.

132 **CSF evidence and published coefficients**

133 CSF evidence was assembled from selected reports rather than a systematic literature review.
134 For the quantitative synthesis, all ten non-IL8 CSF proteins with coefficients and standard
135 errors in both cohorts of Table 4B in Kmezic et al. were eligible, irrespective of effect
136 direction or statistical significance. This corrected an earlier extraction that omitted IL6 and
137 CD5. The source regressions adjusted for age, sex and sample handling. Inverse-variance
138 fixed-effect estimates and normal-approximation 95% confidence intervals were calculated,
139 with Benjamini–Hochberg correction across ten pooled tests. DerSimonian–Laird estimates,
140 Q and I^2 were reported as heterogeneity assessments. The normal approximation does not
141 reproduce the source regressions’ small-sample tests, and pooled significance does not
142 establish significance in both cohorts. IL8, the source study’s separately reported primary
143 biomarker in Table 3, was retained as contextual evidence. The selected table does not contain
144 the full assayed proteome. Published identifications and annotations from PXD002911 and
145 three other reports were recorded in a descriptive evidence map. Counts of reports and

platforms describe coverage of these four studies, not effect magnitude or statistical strength. No participant-level CSF matrix was reconstructed or reanalysed. Figure 3C counts selected reports and assay platforms covering the displayed programs. It is a descriptive tally based on the published evidence table. It is not a systematic meta-analysis of all available CSF proteins, a measure of effect magnitude, or a count of newly analyzed independent cohorts. The Kmezc coefficient synthesis is shown separately in panels A and B. IL8 is reported separately in source Table 3 and was not included in the non-IL8 Table 4B synthesis. Its source-study findings provide context only. The ten-protein correction yields nine positive concordant proteins and one discordant protein, CD5, with pooled $q = 0.984$. Pooled normal-approximation tests must not be substituted for the original cohort-specific regression tests.

Genetic candidate localization

The selection of *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* is based on published GWAS, Mendelian randomization and colocalization results from Du et al. 2024. Those estimates are retained with their source DOI. No independent GWAS, MAGMA analysis, TWAS or new colocalization analysis was performed for this release. Candidate localization and the 5000-resample bootstrap were calculated from donor and cell-group summaries, using the fixed random seed 20260904. The public workbook retains disease-group expression and detection estimates, bootstrap intervals and CIDP versus CIAP contrasts. None of the forty contrasts displayed in Figure 5D survives FDR 0.05. The underlying Genetic_donor_celltype table is omitted from this public workbook. The supplied compilation and bootstrap scripts start from that table and do not reconstruct it from raw matrices.

169 **Published external nerve observations**

170 Stascheit et al. reported C5b-9 deposition in 52 of 55 CIDP nerve biopsies, together with
171 histological and proteomic observations associated with PXD056286. This study calculates
172 the proportion and Wilson interval from that published count. It does not independently
173 examine those biopsies or reanalyse the raw proteome. Supplementary Figure 2 has been
174 redrawn from the retained aggregate values, with source attribution.

175 **Reproduction and figure provenance**

176 Version 41 retains the corrected ten protein CSF synthesis and all five main figures from
177 version 39, with the prose improvements from version 40. It adds Fc sensitivity tests, raw
178 nerve reaggregation and source feasibility audits. Supplementary Figures 3 and 4 are new and
179 are generated from the retained aggregate result tables. Earlier blood and genetic matrix
180 processing was not repeated.

181 Figure 4A retains the previously verified localization-plot interior as a raster asset; its
182 surrounding labels and all other figure text are generated by code. Supplementary Figure 1
183 was redrawn from the retained donor-bootstrap result table and displays all available cell
184 groups. No new donor bootstrap analysis was run. Supplementary Figure 2 was redrawn from
185 published aggregate counts and qualitative observations.

186 Original matrix-processing scripts remain available after downloading the required inputs.

187 Recorded dependency specifications for the earlier pipeline and summary verification are
188 retained with the v2.1.0 archive. Input_files and Data_reconstruction in Supplementary Data 1
189 distinguish public inputs, extracted published results and locally generated intermediate
190 tables.

191 **Data and code access**

192 Repository release: [https://github.com/applepie567/gbs-cidp-blood-nerve-barrier/releases/](https://github.com/applepie567/gbs-cidp-blood-nerve-barrier/releases/tag/v2.3.0)
193 tag/v2.3.0

194 The public archive accompanying manuscript v42 is version 2.3.0 at
195 <https://doi.org/10.5281/zenodo.22641099>. It contains the CSF synthesis, macrophage
196 sensitivity analyses and supplementary data from v41, together with the Human Genomics
197 manuscript and updated figures. The previous v2.2.0 archive is
198 <https://doi.org/10.5281/zenodo.22636223>. Earlier versions remain available through the
199 concept DOI <https://doi.org/10.5281/zenodo.22226672>.
200 Code is provided under the MIT license. The project-derived data tables and figures are
201 covered by LICENSE-DATA. External source materials remain subject to their original terms
202 and attribution requirements.

203 **Original files and local reconstruction**

204 For GSE285983, download GSE285983_RAW.tar and GSE285983_metadata_all.csv.gz
205 using the links in Input_files. Extract the count archive into data/raw/GSE285983/h5/ and run
206 analysis/01_cidp_nerve_pseudobulk.py. This script reads the public donor metadata locally
207 and writes the targeted-expression, module-score and cell-fraction intermediates. The five
208 selected CIDP module scores can then be pivoted into a donor-by-module table for the ten
209 correlation tests.

210 Genetic localization uses nucleus-level $\log_2(\text{CP10k}+1)$ expression summarized by donor and
211 cell group, with detection proportions. The supplied genetic compilation and bootstrap scripts
212 require the precomputed Genetic_donor_celltype table. Exact end-to-end reproduction of that
213 component therefore requires the intermediate table or a separately implemented and
214 validated reconstruction from the original inputs. The public workbook provides the resulting
215 group-level summaries, contrasts and bootstrap intervals.

216 The donor worksheets excluded from the public aggregate workbook remain available for
217 private author use. Missing original acquisition dates from the preceding archive have not
218 been reconstructed. Files newly downloaded on 7 September 2026 have a separate manifest

219 giving their actual acquisition date, source URL, byte count and SHA256 checksum. A source
220 access check is not labelled as a file download.

221 **Revision-specific reproduction**

222 The retained analysis_update directory reproduces the corrected CSF calculations and earlier
223 figures. The strengthening_v41 directory contains strengthen_fc.py, subtype_analysis.py and
224 plot_strengthening.py, together with aggregate results, figure files, the exploratory analysis
225 specification and source manifests. The private checkpoint can reproduce the refined analyses
226 with the reuse checkpoint option. Full reconstruction requires the original matrices and
227 metadata listed in the manifest. The original donor expression table is passed explicitly as the
228 original results directory. No new genetic bootstrap analysis was run.